



Polycystic Ovarian Disease (PCOD): A Review Article

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ABSTRACT:

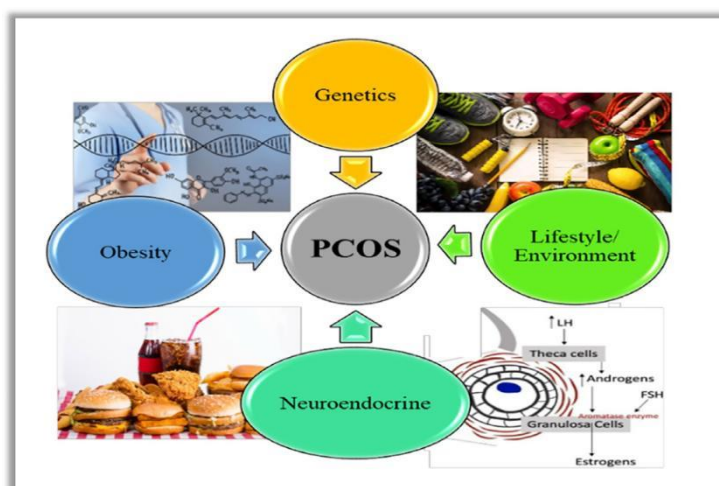
PCOS, or poly-cystic ovarian syndrome, is an endocrine condition. Unlike other ovulation diseases, which occur when the ovaries are defective or non-functioning, this syndrome is characterized by prolonged an-ovulation and ovarian failure. The World Health Organization (WHO) estimates that approximately 116 million women (3.4%) globally suffer from PCOS. The primary pathophysiological factors of PCOS are hyperandrogenism, insulin resistance, and hormonal dysfunction, which decrease folliculogenesis and increase the risk of related comorbidities such as type II diabetes and endometrial cancer. Reducing symptoms of hyperandrogenism, reestablishing monthly regularity, and achieving pregnancy are the main goals of treatment. This review addresses both therapeutic options and potential causes of this elevated risk.

KEYWORD:

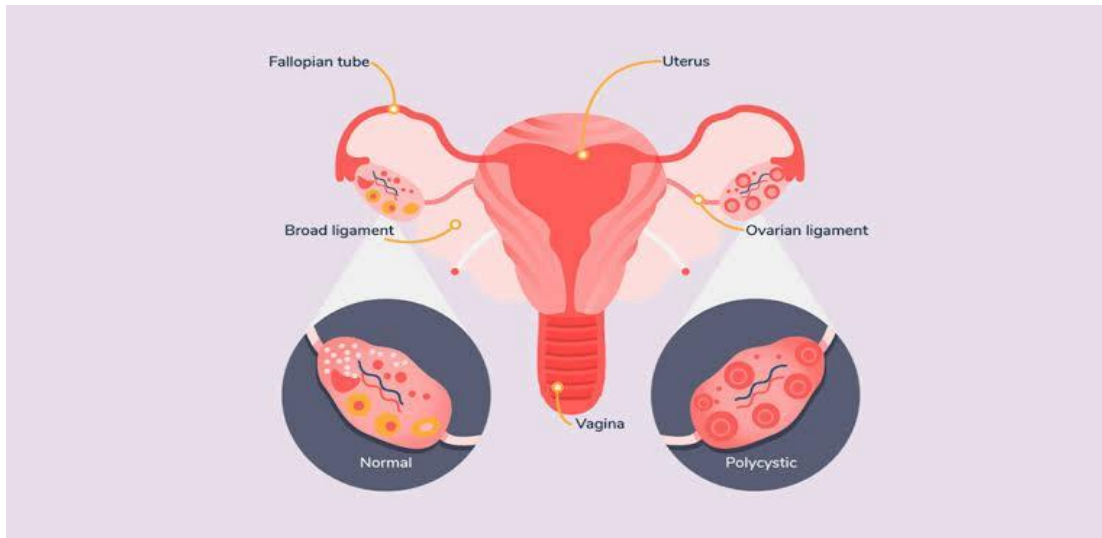
The complication, causes, and treatments of polycystic ovarian syndrome (PCOS) include oral contraceptive pill use, irregular menstruation, hyperandrogenism, and lifestyle modifications.

INTRODUCTION:

The symptoms of polycystic ovarian syndrome (PCOS) include tiny cysts on the ovaries, high testosterone levels, and irregular menstruation cycles. While polycystic ovarian syndrome is primarily a metabolic disorder (hyperandrogenemia), polycystic ovaries are a morphological problem. The PCOS symptoms of hyperandrogenism include ovarian microcysts, anovulation, irregular monthly periods, and inhibition of follicular development. Recently, there has been progress in our comprehension of the pathophysiological process, diagnosis, and treatment of PCOS. We are concentrating on therapy, complications, and lifestyle changes in this study. The selection of diagnostic criteria determines the prevalence of PCOD. As of 2010, the



World Health Organization estimated that 116 million women globally (3.4% of women) were affected. According to community-based prevalence research, around 18% of women had PCOD, and 70% of them had never received a diagnosis. 8–25% of normal women have polycystic ovaries, according to ultrasound results, while 14% of women using oral contraceptives have polycystic ovaries.



Common Symptoms Of PCOD

Loss of hair: Thinning hair on the head is a common symptom of PCOS in women, and this condition may get worse with age.

Acne-prone or oily skin: Due to hormonal fluctuations, PCOS can result in acne and greasy skin. (Of course, you don't need to have PCOS to experience comparable skin problems.)

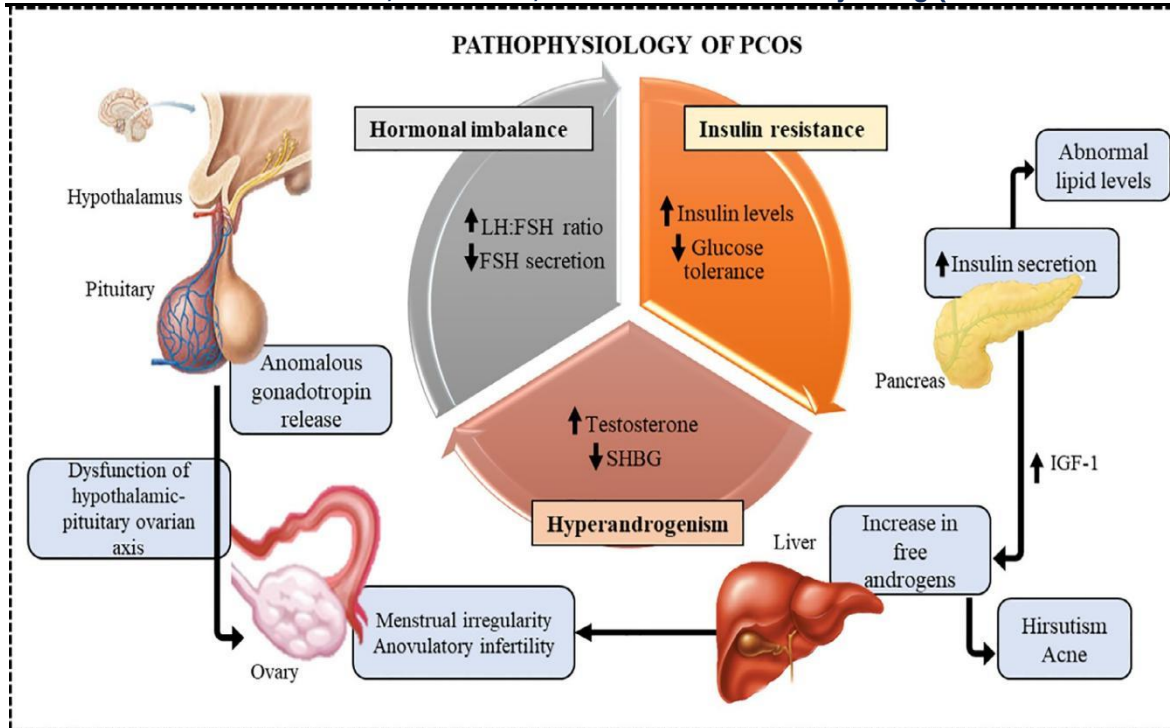
Skin discoloration: You might detect thick, velvety, black patches of skin under your arms or breasts, on the back of your neck, and in the groin area.

Insomnia or persistent tiredness : You can experience trouble falling asleep. It's also possible that you have sleep apnea. This suggests that even if you get adequate sleep, you won't wake up feeling refreshed.

Headaches: Headaches may also be brought on by the elevated hormone levels that lead to PCOS during extended periods of stress. Severe bleeding and irregular periods are two significant menstrual cycle abnormalities that can be brought on by PCOS.uneven time intervals. It's possible that your period won't come on at all or that it will come later than expected.

Inability to conceive: If your cycles are irregular, getting pregnant may be challenging. Among the most frequent causes of infertility is PCOS.

Gaining weigh: Approximately 50% of PCOS - affected women experience weight gain or difficulties losing weight. Gaining a considerable amount of weight is a possible side effect of PCOS. Moreover, PCOS symptoms may worsen if a person is overweight. You can improve your period timing by dropping a few pounds. Losing weight will assist you in maintaining healthy blood sugar levels and decent cholesterol, both of which are essential if you have PCOS.



The prevalence of androgen in PCOS patients is a distinguishing characteristic of this illness.

Hyperandrogenism : Elevated blood levels of free (unbound) testosterone, a crucial hormone influencing the pathophysiology of PCOS, are indicative of the condition. The primary pathophysiological components of this complicated disorder are dissected. Predisposing risk factors for polycystic syndrome development include genetics, neuroendocrine, lifestyle/environment, and obesity. Depending on the population, environmental factors such as food, lifestyle, and physical activity might differ greatly. Endocrine-disrupting substances and glycotoxins are examples of environmental variables that can result in genetic variation, disturbance of the metabolic and reproductive pathways, and the development of PCOS phenotypes and associated comorbidities.

Hormonal imbalance: For women with PCOS, hormonal imbalances are prevalent. Exposure to androgens can disrupt hormone levels, causing the high pulse frequency of GnRH to increase. This can impact the ratio of LH:FSH, causing follicular arrest and dysplasia. Hyperinsulinemia, hyperandrogenism, oxidative stress, irregular periods, and ultimately an increase in the metabolic syndrome are caused by these causes.

Insulin resistance

Hormonal imbalance is common in PCOS-affected females. The hormone levels can be upset by exposure to androgens, which raises the high pulse frequency of GnRH. Follicle arrest and dysplasia may result from this, which may affect the LH:FSH ratio. These factors lead to hyperinsulinemia, hyperandrogenism, oxidative stress, irregular menstruation, and eventually a rise in the metabolic syndrome.

Complications Due To PCOD

Early-term complications : Per the prior definition, one of the primary symptoms linked to PCOS was infertility. An epidemiologic study indicates that PCOS is the most common cause of ovulatory disruption, and oligoanovulation is associated with a higher risk of infertility.

Complications during pregnancy: It has long been believed that ovulatory infertility is linked to PCOS. Currently, particularly in light of the introduction of the updated diagnostic criteria, the focus of attention has shifted to reproductive issues, particularly obstetric challenges. Moreover, the condition's comorbidities—obesity and/or IR—may raise the likelihood of pregnancy.

Long -term complication

• **Cardiovascular complication** : Major scientific bodies report that women with PCOS are more likely to have nontraditional risk factors such as homocysteine, C-reactive protein (CRP), and tumor necrosis factor, as well as classic risk factors like diabetes, obesity, dyslipidemia, and hypertension. PCOS is associated with heightened cardiovascular disease (CVD) risk factors, which can develop even in the absence of obesity but are amplified by body fat at any age.

• **Possibility of metabolic risk**: Obesity is a major global health issue that is spreading rapidly, particularly among children and adolescents. PCOS women are more likely than healthy women to be overweight (BMI >25 kg/m²) or obese (BMI >30 kg/m²), with up to 61% of the population falling into these categories. Even though estimates of obesity in PCOS women differ significantly between nations and ethnic groups, there is a steady central distribution of adipose tissue.

Cancer risk: Since PCOS is a lifelong, multisystemic, and multifaceted disorder, there may be a link between the reproductive and metabolic alterations it brings about and a higher risk of cancer, specifically endometrial, ovarian, and breast cancer, which are diseases that are recognized for their hormonal and/or metabolic pathogenetic processes. A potential pathway that may have facilitated the development of neoplastic diseases in these women, particularly endometrial cancer, is a chronic anovulatory state, which leads to unopposed estrogen activity and is coupled with hyperandrogenism.

Diagnosis

The intrinsic characteristics of polycystic ovarian syndrome, such as the heterogeneity of symptoms and their unpredictability across age groups, make the illness challenging to diagnose. PCOS is characterized by infertility, oligomenorrhea, or amenorrhea, and the existence of cystic ovaries, which are first detected via a laparotomy and then verified by biopsy. A transvaginal scan provides a clearer, more precise picture of the internal anatomy of the ovaries, particularly in obese women. The ultrasonography examination enables evaluation of the ovary's internal and exterior characteristics. One of the most obvious signs of PCOS is excess androgens, or hyperandrogenism, which can be identified by a physical examination or laboratory testing to check for elevated androgen levels in the serum.

Polycystic Ovary Syndrome: Diagnostic Criteria

NIH Criteria (1990)	ESHRE/ASRM Rotterdam Criteria(2003)	Androgen Society (AES) Excess Criteria (2006)
➤ Hyperandrogenism ➤ Chronic anovulation ➤ Exclusion of other endocrinopathies	➤ Hyperandrogenism ➤ Oligo-anovulation ➤ Polycystic ovarian morphology	➤ Hyperandrogenism ➤ Oligo- anovulation ➤ Polycystic ovarian morphology ➤ Exclusion of other related disorders

To identify the precise hormonal imbalance, laboratory measurements of prolactin, dehydroepiandrosterone, thyroid-stimulating hormone, FSH, LH, and testosterone levels can yield additional diagnostic information. HDL and total cholesterol were also measured.

Pharmacological management

Clomiphene citrate

For PCOS patients, it serves as the first-line medication for inducing ovulation. The estrogen receptor antagonist is the one that increases the availability of FSH by interfering with the estrogen signaling pathway's negative feedback loop. Follicle growth is caused by elevated FSH. It includes the initial phase of the menstrual cycle. Infertility is another condition for which it is used.

Metformin

Certain hyperandrogenic indications are counteracted by insulin-sensitizing medications like metformin and troglitazone, which lower the concentrations of free and total testosterone. It also lowers insulin resistance, promotes ovulation, and controls abnormally high testosterone levels.

Glucocorticoids

Insulin-sensitizing drugs such as metformin and troglitazone, which reduce the concentrations of free and total testosterone, counteract several hyperandrogenic symptoms. In addition, it stimulates ovulation, reduces insulin resistance, and regulates excessively elevated testosterone levels.

Gonadotropins

After clomiphene citrate resistance, it is used as a second line of treatment. When FSH₄ is administered under strict supervision and at modest levels initially, it triggers ovulation and maintains and stimulates optimal follicle growth.

N-acetyl-cysteine (NAC)

It contains antioxidants that the body needs to produce glutathione, which reduces oxidative stress and keeps hyperinsulinemia from occurring.

Essential nutrition required in PCOD

Vitamin B6 - The management of PCOD requires this vitamin, which also helps with hormone balance. This vitamin is found in soybeans, Brussels sprouts, spinach, yoghurt, almonds, eggs, and wild rice.

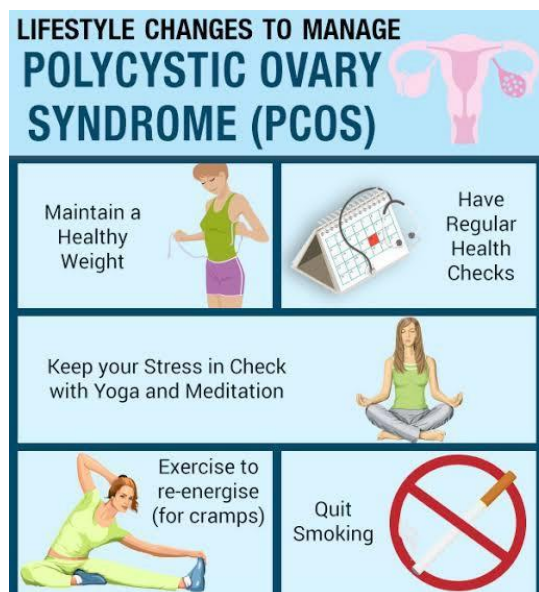
Vitamins B2, B3, B5, and B6 – Are especially helpful in managing weight. present in beans, green veggies, milk, eggs, red meat, and yeast.

Vitamin B5 - This vitamin controls fat metabolism, which helps with weight loss. This nutrient can be found in eggs, meat, beans, avocados, yoghurt, and yogurt.

Vitamin B6 - For adequate thyroid hormone synthesis, is necessary in addition to vitamins B2 and B3. This vitamin can be found in carrots, brown rice, lentils, sunflower seeds, salmon, tuna, chicken, and turkey.

Inositol - Supplemental inositol facilitates insulin signaling. It is unclear how it affects the biochemical and metabolic aspects of PCOS. Ovulation and menstrual cycles may be enhanced, per a recent study. Inositol has a low risk of side effects and is inexpensive; however, this guideline warns against using it due to its limited benefits.

Lifestyle modifications



For individuals troubled by overweight or obesity, changing one's lifestyle is thought to be the first line of treatment. Increased physical activity and exercise, appropriate dietary adjustments, and adherence-maintaining techniques are among the most successful therapies. Additionally, it seems that altering one's lifestyle brings the menstrual cycle and ovulation function closer to normal.

Diet- A large number of PCOS patients develop insulin resistance. High insulin levels are thought to worsen the risk of problems and increase androgens in women with PCOS. It is possible to manage PCOS with a balanced diet.

- 1) Dietary fiber helps combat insulin resistance such as beans, lentils, berries, broccoli, green peas, chia seeds, nuts like almonds.
- 2) Anti-inflammatory foods such as tomatoes, green leafy vegetables, nuts like almonds and walnuts, fruits such as oranges, cherries, blueberries and olive oil.
- 3) Whole grains are high in fiber and won't affect blood sugar as much as refined carbohydrates.
- 4) Fatty acids such as fish oil taken with Vitamin E has also been found to decrease insulin levels.

Exercise: For weight gain prevention, the 2018 PCOS guideline suggests 150 minutes or less of moderate to vigorous intensity exercise per week, and for weight loss and weight regain prevention, 250 minutes or less of moderate to vigorous intensity exercise per week. It's also advised to reduce the amount of time spent sitting down and to include weight training two days a week.

CONCLUSION

With a wide range of clinical manifestations, PCOS is a complicated hormonal, metabolic, and psychological condition. One of the most typical causes of infertility is this. The underlying cause of PCOS is yet unknown. Lifestyle modifications should be the primary therapeutic prescription for PCOS-related infertility before considering any pharmaceutical choices. Although vitamin D and inositol may be beneficial to one's health, more research and guidance are needed. The effectiveness of alternative treatments is not well supported by evidence.

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